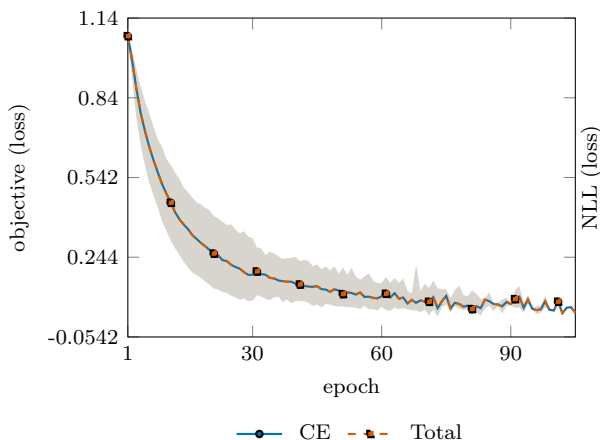
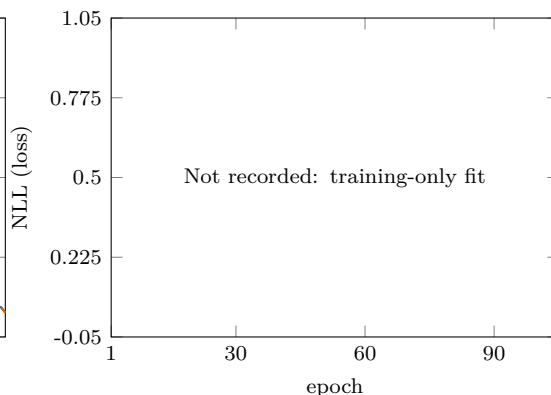


planned=780, monitored=780, missing-history=0, status=complete_history_coverage

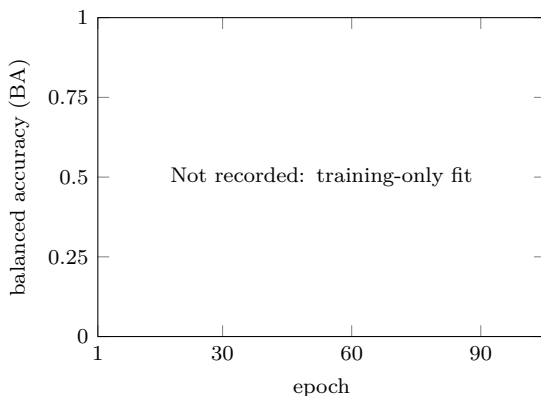
A sampled minibatch objective



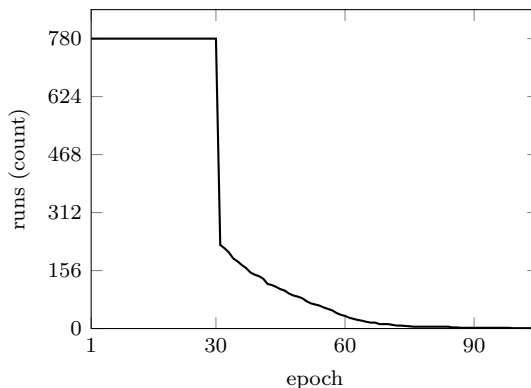
B post-epoch NLL



C balanced accuracy



D contributing fitting runs



RQ-S01 / Scope E: descriptive training diagnostics. 400-1800 cm⁻¹ inputs: SG uses impulse replacement and Savitzky-Golay smoothing; arPLS uses impulse replacement and baseline correction; both finish with [0,1] min-max scaling. CE = chemical cross-entropy; SupCon = supervised contrastive; Paired = paired consistency; NLL = negative log-likelihood; BA = balanced accuracy. Authenticated complete monitor histories aggregated across unique fitting jobs into descriptive policy/recipe/stage curves. Curves summarize per-epoch minibatch-weighted training chemical cross-entropy and the composite total optimization objective; these sampled minibatch objectives differ from post-epoch evaluation negative log-likelihood. Auxiliary supcon/paired loss components are recipe-specific and are not directly comparable to the total objective across recipes. Validation metrics are recorded for source_fit runs only and never substitute for a held-out test set; calibration_model_fit and final_refit rows carry training-only diagnostics. Source_fit optimization uses a 30-epoch minimum stopping threshold and a 200-epoch maximum; refit stages reuse the caller-selected duration. Curves are fitting-run-weighted descriptive summaries over correlated runs (seeds/splits). Lines show medians; 10th/90th percentiles describe variation among saved fitting runs, not confidence intervals. Only runs that reached each epoch contribute; early stopping changes the composition, and stopped runs are not carried forward. Fitting jobs are not independent chemical samples: the dataset has 598 spectra from 69 physical samples and 10 instruments. Coverage is limited to saved SG/arPLS monitor histories; historical MIN losses are not pooled or invented. A missing history is not a zero loss or proof of a failed model. These curves do not support hypothesis tests, superiority claims, model selection, or new uncertainty inference. P08 learned-method performance is reported elsewhere.